



1 Nonproprietary Names

BP: Ammonium Chloride
PhEur: Ammonium Chloride
USP: Ammonium chloride

2 Synonyms

Ammonii chloridum; ammonium muriate; E510; sal ammoniac; salmiac.

3 Chemical Name and CAS Registry Number

Ammonium chloride [12125-02-9]

4 Empirical Formula and Molecular Weight

NH₄Cl 53.49

5 Structural Formula

See Section 4.

6 Functional Category

Acidifying agent; therapeutic agent.

7 Applications in Pharmaceutical Formulation or Technology

Ammonium chloride is used as an acidifying agent in oral formulations. It is also used as a food additive and antiseptic agent.⁽¹⁾

Ammonium chloride is used in the treatment of severe metabolic alkalosis to maintain the urine at an acid pH in the treatment of some urinary tract disorders or in forced acid diuresis.^(2–4) It is also used as an expectorant in cough medicines.⁽⁵⁾

8 Description

Ammonium chloride occurs as colorless, odorless crystals or crystal masses. It is a white, granular powder with a cooling, saline taste. It is hygroscopic and has a tendency to cake.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for ammonium chloride.

Test	PhEur 6.0	USP 32
Identification	+	+
Characters	+	—
Appearance of solution	+	—
Acidity or alkalinity	+	+
Loss on drying	≤ 1.0%	≤ 0.5%
Residue on ignition	—	≤ 0.1%
Thiocyanate	—	+
Bromides and iodides	+	—
Sulfates	≤ 150 ppm	—
Sulfated ash	≤ 0.1%	—
Calcium	≤ 200 ppm	—
Iron	≤ 20 ppm	—
Heavy metals	≤ 10 ppm	≤ 0.001%
Assay (dried basis)	99.0–100.5%	99.5–100.5%

10 Typical Properties

Acidity/alkalinity pH = 4.5–5.5 (5.5% w/w aqueous solutions at 25°C)

Density (bulk) 0.6–0.9 g/cm³

Hygroscopicity Hygroscopic with potential to cake.

Melting point Decomposes at 338°C; sublimes without melting.⁽⁶⁾

Solubility Soluble in water; hydrochloric acid and sodium chloride decrease its solubility in water. Also soluble in glycerin; sparingly soluble in methanol and ethanol. Almost insoluble in acetone, ether, and ethyl acetate.

Specific gravity 1.527 g/cm³

Vapor pressure 133.3 Pa (1 mmHg) at 160°C

11 Stability and Storage Conditions

Ammonium chloride is chemically stable. It decomposes completely at 338°C to form ammonia and hydrochloric acid. Store in airtight containers in a cool, dry place.

12 Incompatibilities

Ammonium chloride is incompatible with strong acids and strong bases. It reacts violently with ammonium nitrate and potassium chlorate, causing fire and explosion hazards. It also attacks copper and its compounds.

13 Method of Manufacture

Ammonium chloride is prepared commercially by reacting ammonia with hydrochloric acid.

14 Safety

Ammonium chloride is used in oral pharmaceutical formulations. The pure form of ammonium chloride is toxic by SC, IV, and IM routes, and moderately toxic by other routes. Potential symptoms of overexposure to fumes are irritation of eyes, skin, respiratory system: cough, dyspnea, and pulmonary sensitization.⁽⁷⁾ Ammonium salts are an irritant to the gastric mucosa and may induce nausea and vomiting.

LD₅₀ (mouse, IP): 1.44 g/kg⁽⁸⁾

LD₅₀ (mouse, oral): 1.3 g/kg

LD₅₀ (rat, IM): 0.03 g/kg⁽⁹⁾

LD₅₀ (rat, oral): 1.65 g/kg⁽¹⁰⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled.

All grades of ammonium chloride must be kept well away from nitrites and nitrates during transport and storage. They must be stored in a dry place, and effluent must not be discharged into the drains without prior treatment.

Ammonium chloride decomposes on heating, producing toxic and irritating fumes (nitrogen oxides, ammonia, and hydrogen chloride).

16 Regulatory Status

GRAS listed. Included in the FDA Inactive Ingredients Database (oral syrup, tablets). Accepted for use as a food additive in Europe. Included in medicines licensed in the UK (eye drops; oral syrup).

17 Related Substances

Ammonia solution.

18 Comments

Ammonium chloride has the ability to cross the red blood cell membrane, and a solution that is isotonic to blood will still cause hemolytic rupture because it acts as a hypotonic solution.

A specification for ammonium chloride is contained in the Food Chemicals Codex (FCC).⁽¹¹⁾

The EINECS number for ammonium chloride is 235-186-4. The PubChem Compound ID (CID) for ammonium chloride is 25517.

19 Specific References

- 1 Gottardi W *et al.* N-Chlorotaurine and ammonium chloride: an antiseptic preparation with strong bactericidal activity. *Int J Pharm* 2007; 335: 32–40.
- 2 Mainzer F. Acid therapy with neutral salts. *Klin Wochenschr* 1927; 6: 1689–1691.
- 3 Portnoff JB *et al.* Control of urine pH and its effect on drug excretion in humans. *J Pharm Sci* 1961; 50: 890.
- 4 Davies HE. Rise in urine pH and in ammonium excretion during a water diuresis. *J Physiol* 1968; 194: 79–80P.
- 5 Coleman W. Expectorant action of ammonium chloride. *Am J Med Sci* 1916; 152: 569–574.
- 6 Zhu RS *et al.* Sublimation of ammonium salts: a mechanism revealed by a first-principles study of the NH₄Cl system. *J Phys Chem* 2007; 111: 13831–13838.

- 7 NIOSH Pocket Guide to Chemical Hazards (DHHS/NIOSH 97-140) 1997; 16.
- 8 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 231.
- 9 Boyd EM, Seymour KGW. Ethylene diamine dihydrochloride. II. Untoward toxic reactions. *Exp Med Surg* 1946; 4: 223–227.
- 10 Smeets P. Ammonium chloride [and water treatment]. *Tribune de l'Eau* 1994; 47(570): 26–29.
- 11 *Food Chemicals Codex*, 6th edn. Bethesda, MD: United States Pharmacopeia, 2008; 46.

20 General References

- Ingham JW. The apparent hydration of ions. III. The densities and viscosities of saturated solutions of ammonium chloride in hydrochloric acid. *J Chem Soc* 1929; 2059–2067.
- Kumaresan R *et al.* Simultaneous heat and mass transfer studies in drying of ammonium chloride in fluidized bed dryer. *Process Plant Eng* 2007; 25(3): 60–66.

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22 Date of Revision

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Ascorbic Acid

1 Nonproprietary Names

BP: Ascorbic Acid
JP: Ascorbic Acid
PhEur: Ascorbic Acid
USP: Ascorbic Acid

2 Synonyms

Acidum ascorbicum; C-97; cevitamic acid; 2,3-didehydro-L-threo-hexono-1,4-lactone; E300; 3-oxo-L-gulofuranolactone, enol form; vitamin C.

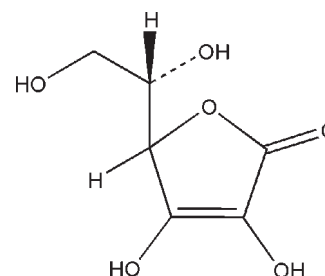
3 Chemical Name and CAS Registry Number

L-(+)-Ascorbic acid [50-81-7]

4 Empirical Formula and Molecular Weight

C₆H₈O₆ 176.13

5 Structural Formula



6 Functional Category

Antioxidant; therapeutic agent.

7 Applications in Pharmaceutical Formulation or Technology

Ascorbic acid is used as an antioxidant in aqueous pharmaceutical formulations at a concentration of 0.01–0.1% w/v. Ascorbic acid has been used to adjust the pH of solutions for injection, and as an adjunct for oral liquids. It is also widely used in foods as an antioxidant. Ascorbic acid has also proven useful as a stabilizing agent in mixed micelles containing tetrazepam.⁽¹⁾